



Society of Petroleum Engineers

SPE-227093-MS

Where are the Fractures? Incorporating Pressure Transient Analysis in Naturally Fractured Carbonate Reservoirs

E. A. Ekpiri, F. Moid, R. Almahdi, and M. Alammari, Saudi Aramco, Dharhan, Eastern Province, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227093-MS](https://doi.org/10.2118/227093-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Naturally fractured carbonate reservoirs (NFCRs) are among the most challenging hydrocarbon systems to characterize and develop due to their dual-porosity nature, complex fracture networks, and highly heterogeneous properties. Pressure transient analysis (PTA) has emerged as an indispensable tool for understanding the unique flow dynamics in these reservoirs, offering critical insights into fracture-matrix interactions, reservoir connectivity, and fluid flow behavior. This paper presents a comprehensive framework for incorporating PTA into the evaluation and management of NFCRs, with a focus on maximizing hydrocarbon recovery and reducing development uncertainties.

By leveraging diagnostic plots, specialized type curves, and advanced analytical methods, PTA is used to unravel complex flow regimes that evolve over time, including early-time fracture-dominated flow, transition periods governed by matrix-fracture exchange, and late-time boundary-dominated behavior. These interpretations are essential for accurately defining reservoir properties and ensuring optimal field development strategies. The tool is used for identification and quantification of essential reservoir characteristics, including fracture permeability, matrix porosity, and key dual-porosity parameters such as the inter-porosity flow coefficient (λ) and the storativity ratio (ω).

The Field applications of PTA in NFCRs highlight its capability to address critical challenges such as identifying reservoir compartmentalization, estimating effective drainage areas, and delineating fracture connectivity. Additionally, this analysis sheds light on key operational hurdles, including fluid losses, significant wellbore storage effects, and unstable pressure responses. Advanced test designs, such as multi-rate or pulse testing, combined with numerical simulations, are proposed to enhance the reliability of PTA results in complex carbonate settings. The integration of PTA with complementary methods such as rate transient analysis (RTA), material balance, and full-field reservoir simulation is emphasized as a best practice.

The presence of the V-shaped feature in the derivative plot of the crestal "well A" but its absence in the flank wells "B and C", despite all wells being located in the same dry gas fractured carbonate reservoir, can be attributed to several factors. Variation in fracture density and orientation across the reservoir can lead to differences in permeability, with the crestal area potentially having a higher fracture density or more

favorably oriented fractures. Reservoir heterogeneity also plays a role, as fractured carbonate reservoirs are inherently heterogeneous, and the crestal well might be located in a more heterogeneous part of the reservoir.

This multidisciplinary approach offers a holistic understanding of NFCR behavior, improving decision-making for well placement, stimulation design, and production optimization. This paper underscores the transformative potential of PTA in naturally fractured carbonate reservoirs, advocating for its inclusion as a core element of reservoir evaluation workflows.

Introduction

Naturally fractured carbonate reservoirs (NFCRs) are among the most prolific yet complex hydrocarbon reservoirs globally, contributing significantly to the world's oil and gas supply. Their geological complexity stems from varied depositional environments, post-depositional diagenetic alterations, and tectonic activities, all of which lead to highly heterogeneous pore structures and the development of intricate fracture networks. These fractures may either enhance fluid flow or compartmentalize the reservoir, depending on their connectivity and orientation.

Understanding the interplay between the porous matrix and the fracture system is crucial for optimizing reservoir management (Figure 1). While the matrix typically stores the majority of hydrocarbons, the fractures often provide the primary pathways for fluid flow. The connectivity, orientation, and properties of these fractures substantially influence well productivity and overall reservoir performance.

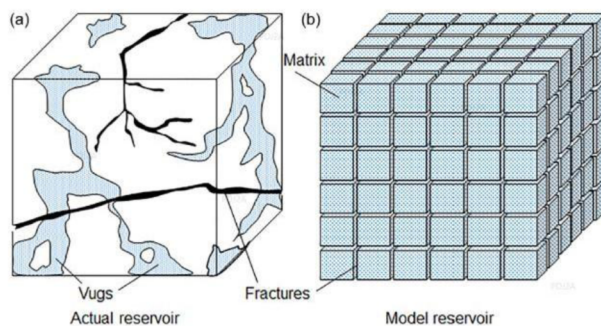


Figure 1—Illustration of a naturally fractured reservoir and its idealized representation in a reservoir model (after Warren and Root 1963).

Pressure Transient Analysis (PTA) offers a non-invasive technique to diagnose the subsurface conditions by interpreting the pressure response of a reservoir to changes in flow rate. Among fractured reservoirs, especially those best described by dual-porosity or dual-permeability models, a distinctive V-shaped signature in the pressure derivative plot is a critical diagnostic feature. This V-shape results from the transition between different flow regimes—initial fracture-dominated flow, inter-porosity flow from the matrix, and finally boundary-dominated flow.

Recognizing and correctly interpreting this V-shape enables reservoir engineers to identify the presence, extent, and connectivity of fractures, facilitating more accurate reserve estimations and development plans. Conversely, homogeneous single-porosity systems typically lack this characteristic V-shaped response, displaying simpler derivative behavior.

Several methods can be employed to identify naturally occurring fractures in gas carbonate reservoirs, each with its strengths and limitations. A comprehensive approach that integrates multiple disciplines and data sources is essential for obtaining a robust understanding of fracture distribution, orientation, and properties. Some of the key methods used for fracture identification include:

- **Borehole Image Logs:** Borehole image logs, such as Formation MicroImager (FMI) and Borehole Televier (BHTV), provide high-resolution images of the borehole wall, allowing for the direct

observation of fractures. These logs can help identify fracture orientation, density, and aperture, as well as distinguish between natural and induced fractures (Figure 2 and 7).

- **Core Data:** Core samples provide valuable information about the reservoir's geological and petrophysical properties, including fracture characteristics. Core analysis can help identify fracture types, such as extensional or shear fractures, and provide insights into fracture filling and sealing (Figure 3).
- **Outcrop Studies:** Outcrop studies involve the examination of exposed rock formations, which can provide valuable insights into fracture patterns, orientation, and distribution. Outcrops can be used to calibrate and validate fracture models, as well as to identify regional fracture trends.
- **Seismic Data:** Seismic data, including 2D and 3D seismic surveys, can be used to identify large-scale fractures and fracture corridors. Seismic attributes, such as amplitude, phase, and azimuth, can help detect fracture-related anomalies and predict fracture density and orientation.
- **Production Performance:** Production data, including flow rates, pressures, and water cut, can be used to infer the presence of fractures. For example, high flow rates or pressure anomalies may indicate the presence of a fracture network.
- **Rate Transient Analysis (RTA):** RTA involves the analysis of production rates and pressures to understand reservoir behavior and identify fracture-related effects, such as fracture-dominated flow or compartmentalization.
- **Pressure Transient Analysis (PTA):** PTA involves the analysis of pressure data to understand reservoir properties, including fracture-related effects, such as fracture storage and transmissivity.
- **X-Ray Diffraction (XRD) and Scanning Electron Microscopy (SEM):** XRD and SEM are laboratory techniques used to analyze core samples and identify mineralogical and textural properties, including fracture filling and sealing.
- **Drilling Losses:** Drilling losses, including mud losses and gas kicks, can indicate the presence of fractures. Drilling parameters, such as rate of penetration and torque, can also be used to infer fracture presence and orientation.

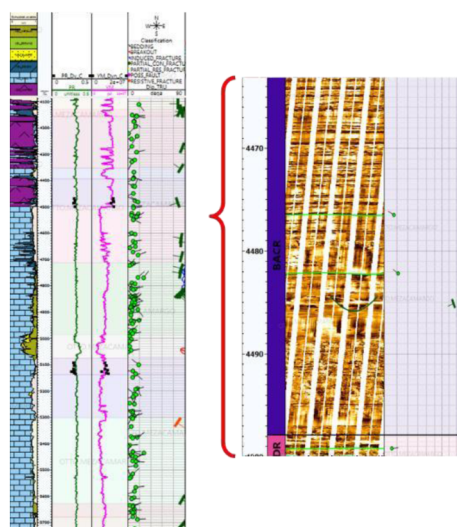


Figure 2—Borehole Images Showing Naturally Occurring Fractures in Carbonates



Figure 3—Core Data Showing Naturally Occurring Fractures in Carbonates

This paper presents a comprehensive framework for incorporating PTA into the evaluation and management of NFCRs, with a focus on maximizing hydrocarbon recovery and reducing development uncertainties. The paper also investigates the origins and implications of the V-shape in derivative plots, elaborates on the theoretical underpinnings of dual-porosity and dual-permeability models, and reviews how these models are employed to interpret PTA data in carbonate reservoirs. Through a comprehensive literature review and detailed discussion, this study provides deeper insights into using PTA for fracture characterization in complex carbonate systems.

Literature Review

The foundation of pressure transient analysis (PTA) in fractured reservoirs can be traced back to the seminal work of Warren and Root (1963), who introduced the dual-porosity model. They conceptualized reservoirs as systems composed of low-permeability matrix blocks interspersed with high-permeability fractures (Figure 4), providing an analytical framework that explained the delayed pressure responses often observed in well tests. Central to their model was the Warren-Root shape factor, which quantified the fluid exchange between matrix and fractures.

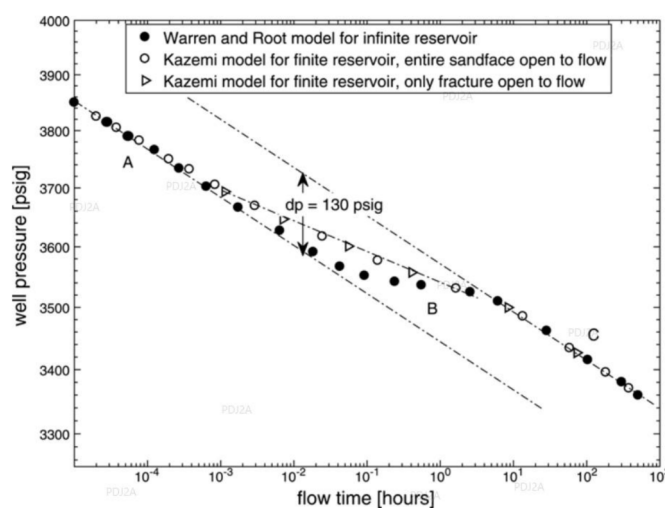


Figure 4—Reproduced the theoretical results of Warren & Roots. On a semi-log plot shows 2 straight lines. Warren and Root (1963)

Warren and Root introduced the first technique for identification and interpretation of natural fracture reservoir. They identified two parameters that control deviation from homogenous reservoir behaviors are:

$$\text{Storativity}(\omega) = \frac{\omega_f c_f}{\omega_f c_f + \omega_m c_m} \quad (1)$$

Storativity defines the ratio of fluid stored in fracture system to that of the total reservoir system. ω_f = fracture porosity, ω_m = matrix porosity, C_f = fracture compressibility, C_m = matrix compressibility

$$\text{Interporosity Flow Coefficient, } IFC(\lambda) = \alpha r_w^2 \frac{k_m}{k_f} \quad (2)$$

IFC measures the ability of the fluid to flow from the matrix into the fractures. r_w = well radius, α = shape factor that depends on the size and geometry of the matrix, k_f = fracture permeability, k_m = matrix permeability. These equations form the backbone of the type curve matching and derivative interpretation processes utilized in this study to characterize the fracture network and matrix contribution to overall flow.

Building upon this, Kazemi (1969) advanced the understanding by developing the dual-permeability model. Unlike Warren and Root's approach, which assumed instantaneous equilibrium, Kazemi's model allowed for transient matrix-to-fracture fluid flow and finite permeability in the fractures. This improved representation of pressure diffusion behavior offered more accurate interpretations of well tests and became a new standard in analyzing naturally fractured systems.

To translate these theoretical advancements into practical applications, Cinco-Ley and Samaniego (1981) developed specialized type curves for fractured wells. They introduced the concepts of bilinear and linear flow regimes, enhancing engineers' ability to diagnose fractures and estimate properties such as fracture conductivity and half-length with greater precision.

Building on these diagnostic improvements, Gringarten (1982) significantly refined the interpretation of well tests by introducing the systematic use of pressure and pressure derivative plots (Figure 5 and 6). These plots enhanced the resolution of flow regimes, particularly in heterogeneous reservoirs, and popularized the V-shaped derivative curve as a diagnostic hallmark of dual-porosity behavior. His work became a cornerstone for modern PTA practices.

Further advancing the clarity of diagnostic methods, Bourdet et al. (1983) introduced a new set of type curves based on pressure and derivative plots. These derivative-based plots improved the identification of flow regimes such as wellbore storage, skin effects, and boundary-dominated flow, becoming foundational tools for modern PTA analysis.

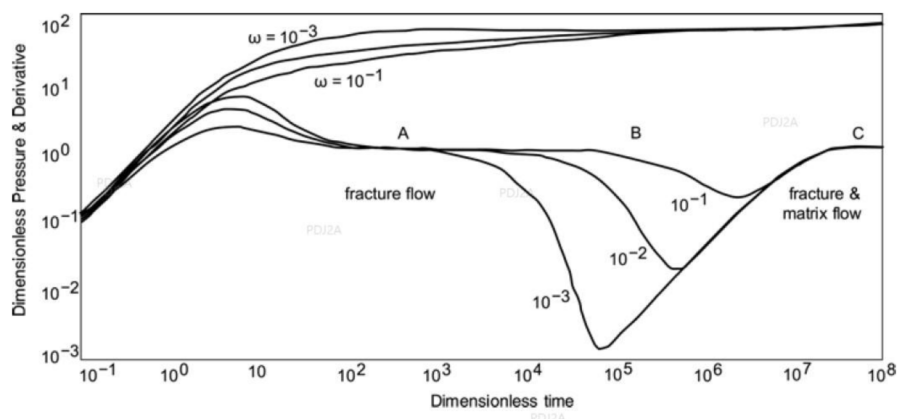


Figure 5—Classical V-Shape on a log-log plot Samaniego (1981)

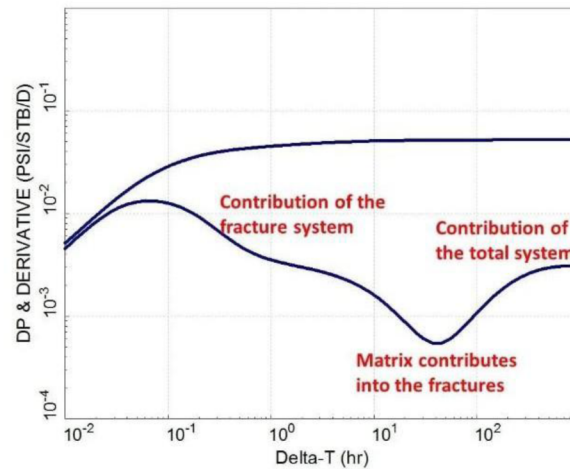


Figure 6—Classical V-Shape on a log-log plot

Recognizing the need for a broader diagnostic perspective, Serra (1984) emphasized integrating well-logging data with pressure analysis. By combining lithological, porosity, and saturation information from well logs with dynamic pressure data, Serra advocated a multidisciplinary approach that improved fracture detection and reservoir evaluations, especially in complex carbonate settings.

Building on the integration of multiple data sources, Tiab and Donaldson (1996) highlighted the importance of petrophysical measurements in calibrating dual-porosity and dual-permeability models. By linking porosity, permeability, and capillary pressure data to flow dynamics, they enhanced the robustness of reservoir models, particularly for fractured systems.

As computational capabilities evolved, Horne (1995) championed the integration of computer-aided methods with traditional analytical techniques. His framework enabled automated generation of diagnostic plots and reduced human error, allowing more efficient and accurate interpretation of complex fractured reservoirs.

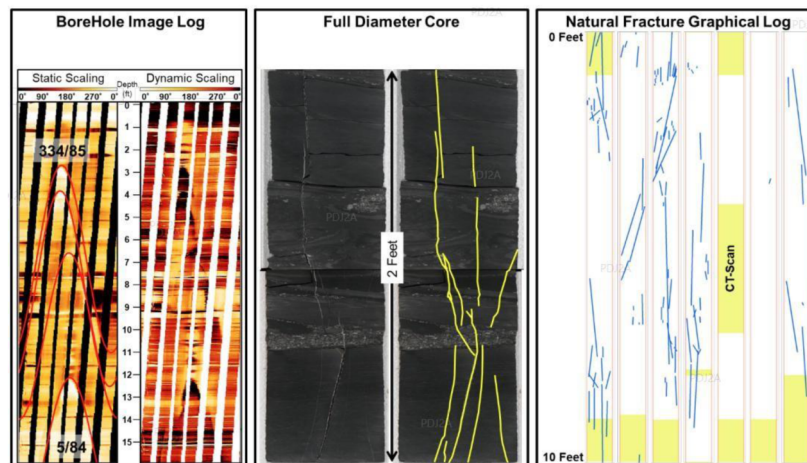


Figure 7—Core Natural fracture characterization. Two left-most figures show 16 feet of BHI log with static (left) and dynamic (right) scaling. Four natural fractures are interpreted in the BHI with static scaling (left), with two orientations displayed. Two figures in the middle shows the same 2 feet of core uninterpreted (left) and interpreted natural fractures (right). Right-most image shows 70 feet of graphical log (natural fractures depicted as blue lines). Yellow areas in right-most figure correspond to core that was preserved and therefore not available for examination; for those core sections CT-Scan images were used to interpret natural fracture presence (D. Gala, 2023).

Expanding the theoretical base, Raghavan (1993) offered a rigorous mathematical treatment of transient pressure behavior, particularly the transition from fracture-dominated to matrix-supported flow. His work

enhanced the interpretation of dual-porosity systems by linking theoretical insights to practical reservoir analysis.

These advancements converged with the practical field-oriented approach of Economides (1995), who linked well test interpretations directly to reservoir stimulation strategies. His work demonstrated how detailed fracture characterization could lead to optimized fracture designs and improved reservoir performance.

Specifically targeting carbonate reservoirs, Gauthier et al. (2002) emphasized the integration of core, log, and PTA data to accurately characterize complex fracture systems. Their holistic approach was critical in understanding scale-dependent permeability and fracture connectivity in heterogeneous carbonates.

Enhancing numerical approaches, Tezcan (2001) introduced improved models for estimating fracture parameters by accounting for complex geometries and anisotropic behavior. His advancements strengthened the diagnosis of fracture network scale and connectivity through PTA responses.

Responding to the need for dynamic modeling, Ghedan (2009) combined pressure transient data with reservoir simulation, capturing the evolving behavior of fractures over time. His simulation-assisted PTA techniques enabled more realistic and field-validated fracture characterizations.

Refining dual-porosity interpretations, Zhao et al. (2011) incorporated heterogeneous matrix block sizes and complex fracture geometries, resulting in more accurate diagnoses of inter-porosity flow behavior and matrix permeability.

Addressing early-time interpretation challenges, Da Prat (1990) developed diagnostic procedures that combined type curve matching with derivative analysis to account for wellbore storage and skin effects. His practical methodology improved the reliability of fracture and matrix parameter estimations in fractured reservoirs. Extending these advancements, Clarkson and Pedersen (2004) applied modern numerical methods to tight gas and fractured reservoirs, enhancing the understanding of fracture networks' impacts on flow and production.

Further improving analytical modeling, Zhou and Gringarten (2002) developed refined models that more accurately captured transitions between flow regimes, providing powerful diagnostic tools for interpreting complex well tests. Field applications by Al-Awad and Al-Mohanna (2009) emphasized the role of fracture anisotropy and heterogeneity in pressure responses. They demonstrated through case studies the necessity of incorporating anisotropic models and advanced diagnostics for more accurate characterization of fractured carbonate reservoirs. Pushing model sophistication further, Kamari et al. (2016) introduced hybrid analytical-numerical approaches that bridged the gap between analytical solutions and full-scale simulations, significantly improving prediction and interpretation of dual-porosity systems.

Finally, looking toward the future, Bahadori (2015) proposed integrating machine learning with traditional PTA techniques. His work suggested that data-driven approaches could further enhance fracture detection and reservoir characterization in increasingly complex environments. Together, these contributions form a coherent and evolving framework, refining pressure transient analysis into a critical tool for diagnosing, modeling, and optimizing production from complex fractured carbonate reservoirs

Methodology

The field under study is a dry gas carbonate reservoir with minimal water production and no H₂S content. The discovery well (Well A) is located at the structural crest, while four appraisal wells (Wells B–E) were drilled along the north, south, east, and west flanks.

To ensure a robust characterization of the reservoir, an extended well testing (EWT) campaign was executed across all three wells. The data acquisition plan was carefully designed to capture both short-term and long-term pressure responses, necessary for identifying complex flow regimes such as dual-porosity behavior. A 45-day extended build-up test was performed on the crestal well (Well A) with ample time to capture matrix-to-fracture transfer and late-time radial flow behavior. The other flank wells (Well B–E) were

shut-in for 10–20 days. While sufficient for capturing early-time radial flow and potential boundary effects, this shorter duration posed a limitation in fully resolving the fracture-matrix interaction.

High-resolution quartz pressure gauges were deployed in each well to ensure precise data acquisition. These gauges are known for their stability, low drift, and ability to detect subtle pressure changes over long periods—an essential feature when diagnosing low-flow matrix contributions in dual-porosity systems.

Pressure transient analysis techniques were applied on the data from the EWT. Diagnostic plotting of log-log pressure and derivative plots were generated to visually identify flow regimes; pressure and derivative data were matched against standard type curves for both dual-porosity (Warren and Root model) and homogeneous reservoirs. The derivative analysis for fracture identification with a special focus placed on analyzing the pressure derivative response. In dual-porosity systems, a characteristic dip (V-shape) in the derivative plot signals the transition from fracture-dominated to matrix-dominated flow.

In addition, data pre-processing involved de-noising and superposition adjustments to correct for variable flow rates during drawdown periods. The quality control of the data ensured that the final analysis could reliably distinguish between flow regimes and fracture contributions.

Results and Observation

Observed early and clear radial flow regime. Some have intermediate V-shaped dip in the pressure derivative while others show absence of pronounced V-shaped dip. Also observed late-time radial stabilization in some wells, while others have transition to boundary effects in some wells.

PTA on Crestal Well A

The well was perforated at the reservoir interval using 4 ½" tubing-conveyed perforating (TCP) guns and recorded an under-balance pressure of 2000 psi. It was initially opened on 32/64" for initial flow and observed gas at surface after 24 minutes (Figure 8). After 44 minutes with almost dry gas at surface, decision made to have initial build up (BU). The final recorded pressure after 2 hrs of downhole shut-in was 8,990 psia. During main flow, the well was opened on 3-choke sizes and the results is shown below. The final recorded pressure after 36 hrs of downhole shut-in was 8,990 psia. A 1 hr initial flow and 2 hrs initial BU was recorded and 20 hrs main flow and 36 hrs final BU recorded.

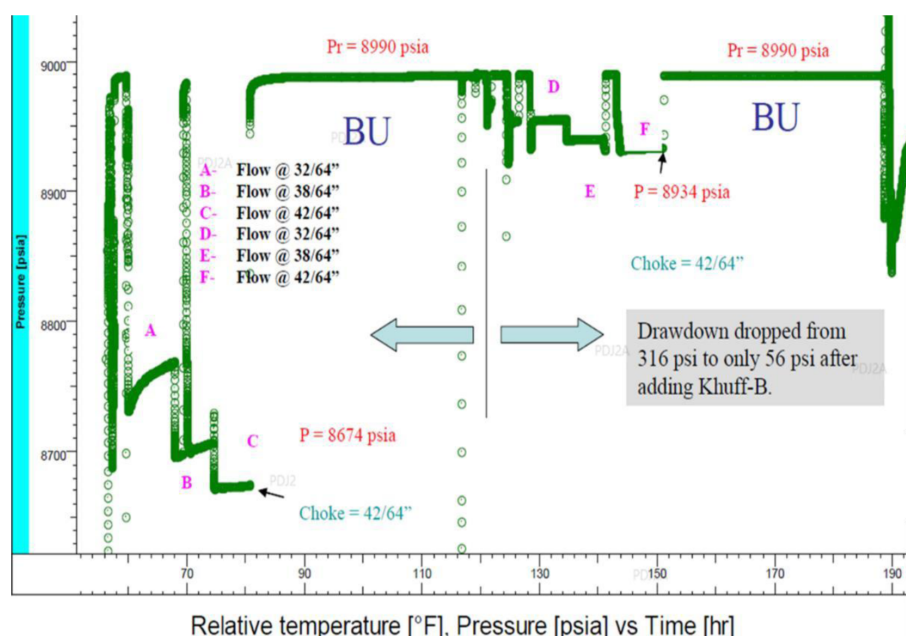


Figure 8—Cartesian plot for the Downhole data for Well A.

Typical in naturally fractured reservoirs, we observe an early-time flat derivative followed by a dip (inter-porosity flow) and then a rise. The initial flat derivative refers to the early-time radial flow in the fracture network while in the dip, fluid begins transferring from the matrix to the fractures (inter-porosity flow) and then finally in the rise; the matrix dominates the response as it equilibrates, leading to a V shape.

The log-log plot for the reservoir alone shows radial composite behavior. Flow capacity is estimated as 25,000 md.ft and increased to 85,000 md.ft (Figure-9). After 10 hours, it could be interpreted slightly that a V-shaped feature on a derivative plot, often referred to as a "V-shaped signature" or "hump" is starting to develop but prematurely curtailed due to the duration of the buildup. The feature typically indicates the presence of a fracture or a high-permeability streak in the reservoir. It is usually more pronounced in the derivative curve because it highlights changes in the slope of the pressure transient response, which can be indicative of different flow regimes. The dual KH estimation also supports two systems. The significant difference between early KH (25,000 md-ft) and late KH (85,000 md-ft) indicates that the initial pressure behavior underestimates the full permeability due to late matrix support. This is a hallmark of fractured system.

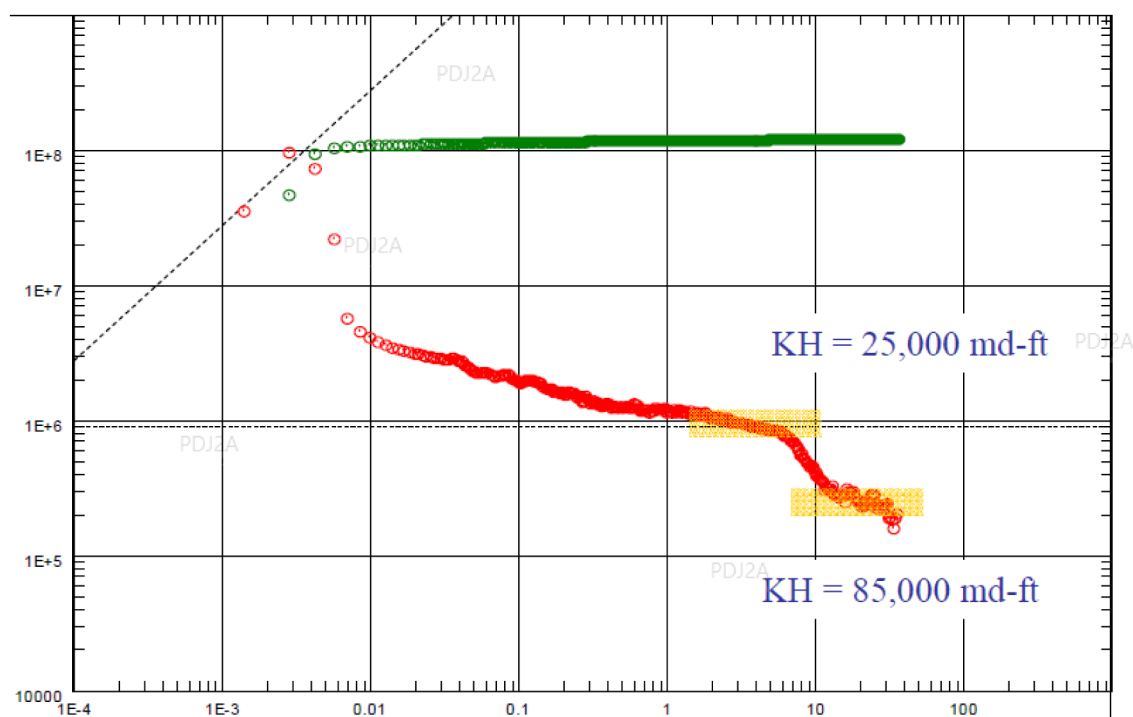


Figure 9—Derivative plot for Well A final Build Up for Well A.

PTA on Flanked Well B

After gun was fired, wellhead pressure (WHP) increase to 3,044 psia. Well was opened to flow on 20/64" fixed choke to the gauge tank for cushion recovery for 1 hr. Well was shut in at surface for initial build up for 1.25 hrs. After initial build up was completed, well was opened on choke 24/64" fixed for 3 hrs, then Choke was increased to 32/64" fixed for 6 hrs. Then well was shut in at downhole valve for final build up for 12 hrs.

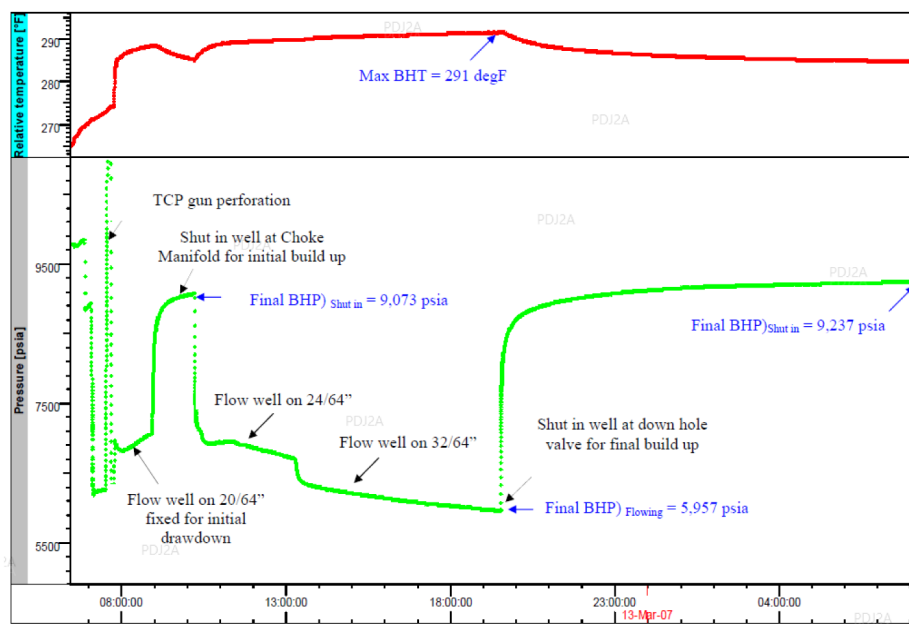


Figure 10—Cartesian plot for the Downhole data *Well B*.

Figure 11 shows the pressure build-up response for a gas well, with the pressure (log-log) and its derivative plotted against the shut-in time. The key question is whether this plot exhibits a V-shape derivative response characteristic of a naturally fractured reservoir (dual porosity). The derivative curve initially rises and forms a broad peak. This is not a sharp V-dip but rather a rounded hump, which may indicate wellbore storage followed by early fracture-dominated flow. We also observe that the derivative stabilizes into a bear- horizontal trend at later times; suggestive of radial flow in the combined matrix-fractured system. This plateau region is typically used to estimate KH. In this plot, there is no clear dip-then-rise transition. Instead the derivative rises then flattens. Overall, this plot does not provide behavior typically associated with fractured reservoirs.

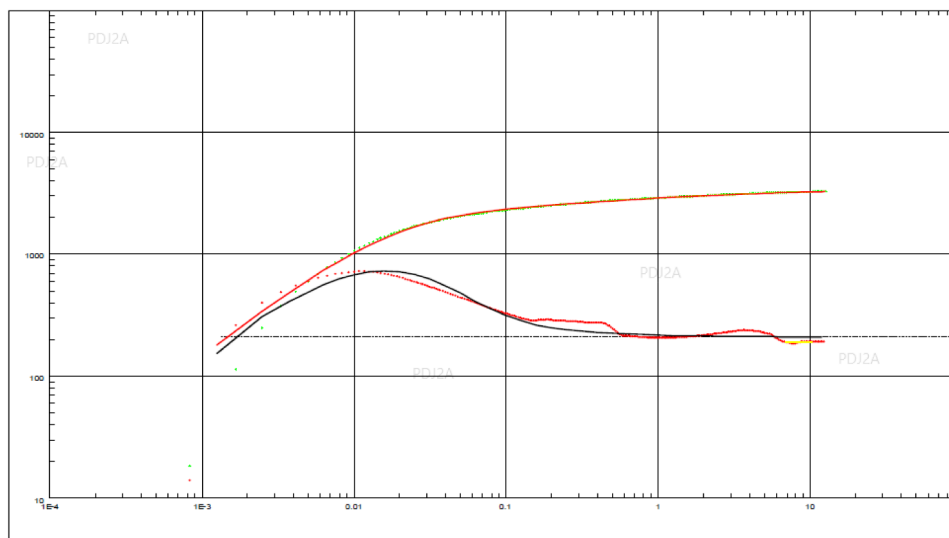


Figure 11—Log-log and derivative plot for final build up for *Well B*

PTA on Flanked Well C

The well was perforated at the reservoir interval and after initial drawdown and build up, the final fall off WHP after well stimulation was recorded as 3,639 psia. The well was opened 40/64" fixed choke and flowed naturally. Well was shut in at down hole valve for 30 hours final build up (Figure 12).

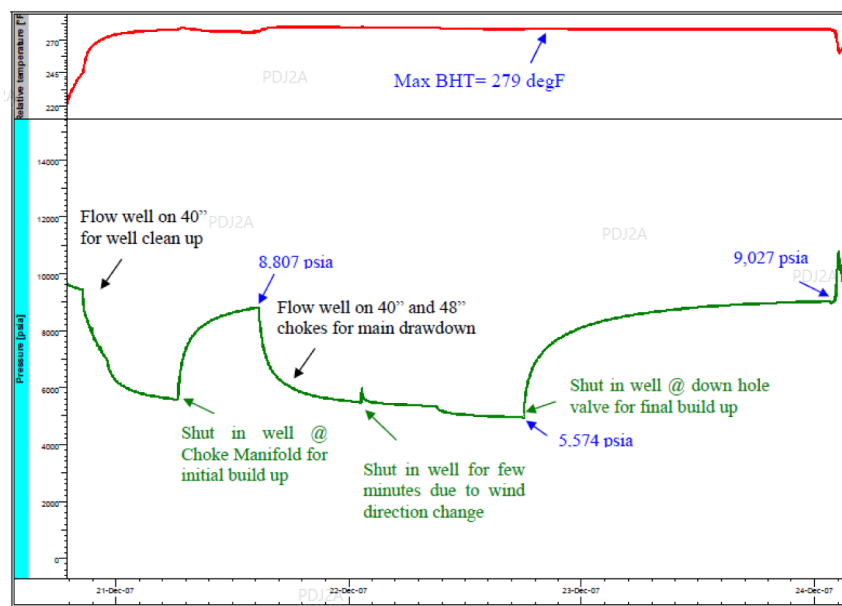


Figure 12—Cartesian plot for the Downhole data for Well C.

The post-stimulation final build up (Figure 13) analysis is showing the response of stimulated tight formation with negative skin. Radial flow at middle time region is showing the reservoir flow capacity. Late time region is showing that reservoir quality is continuously improving. Based on the build up, the flow capacity (kh) is 42.2 md.ft and Skin is -3.7 .

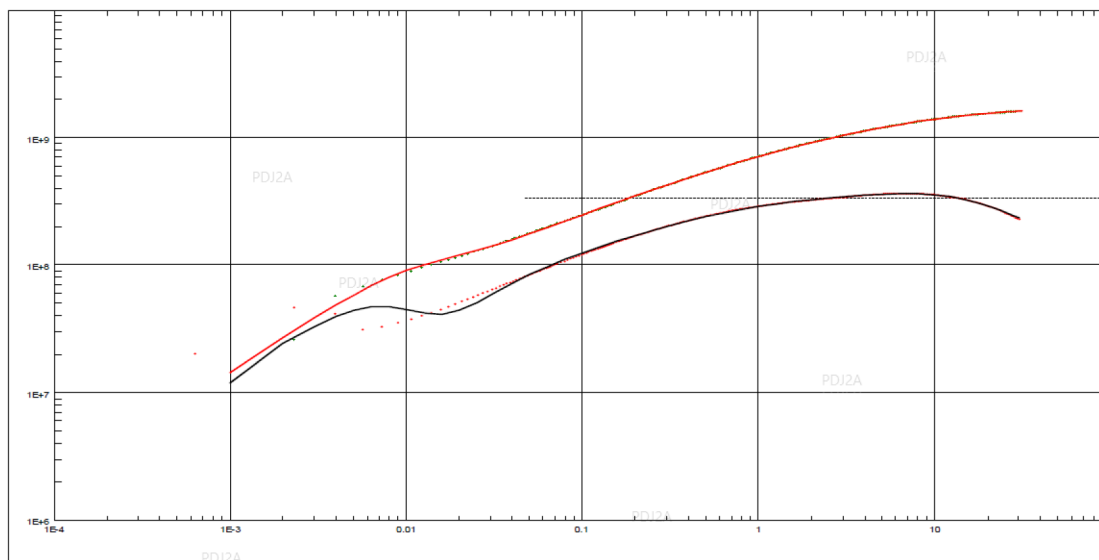


Figure 13—Log-log and derivative plot for final build up for Well C.

The derivative rises gradually, peaks and then flattens at late time. There is no characteristics dip-then-rise pattern (i.e. no "V") that would suggest active dual-porosity flow. We observe good match with

infinite conductivity fracture model. The model used is "infinite conductivity, changing storage (Hegeman)", indicating that the flow is dominated by a stimulated fracture with good conductivity. The derivative matches the expected model behavior without dual-porosity inflection. The late time derivative trend flattens, consistent with radial flow in the stimulated system, possibly after the fracture and formation reach equilibrium. Overall, the derivative plot does not exhibit a V-shape signature. Instead it reflects a flow system dominated by a high-conductivity fracture with minimal matrix contribution over the test duration. This behavior is typical of post-fracture stimulation test, where fluid quickly moves through the created fracture system suppressing the matrix-fracture transition response

Result Discussion

The contrast between the crestal and flank well PTA results underscores the importance of extended shut-in durations in naturally fractured carbonates. The V-shaped derivative signature observed in the crest well points to an active fracture network at the crest, potentially connected to regional fracture swarms. The crest, being structurally elevated, often experiences higher natural fracture density due to tensile stresses and diagenetic influences that promote fracture development. This is consistent with the observation of a distinct dual-porosity flow regime, where fluid initially moves rapidly through fractures and subsequently interacts with the lower-permeability matrix blocks.

In contrast, the flank wells, though lacking clear dual-porosity signals, are not necessarily devoid of fractures. The absence of a V-shaped dip in the derivative plot could largely be attributed to the shorter shut-in durations (10–20 days) employed during the extended well tests. In dual-porosity systems, the matrix-fracture exchange typically requires longer shut-in periods to become apparent in pressure derivative responses. Without sufficient time for the pressure to propagate into the matrix blocks and reveal the inter-porosity flow effects, the diagnostic plot may resemble that of a homogeneous reservoir, even when fractures are present.

Additionally, structural and depositional heterogeneities across the field can lead to spatial variations in fracture intensity and connectivity. Flank areas might inherently have fewer open fractures due to differences in stress regimes and lithofacies distribution, compounding the difficulty in detecting dual-porosity behavior without prolonged observation.

It is also important to consider the impact of gauge resolution and data noise. Subtle deviations indicative of matrix-fracture interactions could be masked in shorter build-ups by pressure noise or insufficient data sampling resolution. Furthermore, rate fluctuations prior to shut-in can introduce superposition effects that obscure the characteristic flow signatures.

These findings reinforce that extended build-up periods (>30 days) are recommended for future appraisal wells, especially in flank regions where fractures may be less pervasive or less connected. Longer shut-in times enable a more complete diagnostic of flow regimes, capturing both early-time fracture-dominated flow and the delayed matrix contribution essential for dual-porosity interpretation.

A comprehensive testing strategy should combine extended shut-in durations with high-quality pressure gauges and stable flow conditions to minimize noise and superposition effects. Such an approach will yield more reliable interpretations, facilitating accurate reservoir characterization and informing optimal field development plans.

Implications for Field Development

Well Placement: The results of the PTA indicate that the crestal area exhibits a well-developed fracture network with active matrix contribution, as evidenced by the V-shaped derivative feature. Prioritizing well placement in the crestal and near-crestal zones will maximize initial gas recovery and production rates by tapping into both fracture and matrix systems. Wells drilled in these areas are expected to benefit from enhanced deliverability and better connectivity, reducing the number of development wells required

to achieve production targets. Additionally, focusing on these zones minimizes the risk of early water breakthrough if a peripheral aquifer is present.

Test Design: The findings highlight the necessity of designing future appraisal and development well tests with extended shut-in durations, preferably 40 days or longer. Long-duration tests are essential to fully characterize the dual-porosity behavior and to accurately delineate matrix and fracture properties. For flank and structurally complex areas where fracture presence is uncertain, extended build-up tests will improve the confidence in flow regime interpretation and fracture mapping. Well test programs should also incorporate pressure monitoring over the entire shut-in period using high-stability quartz gauges to maintain data integrity.

Reservoir Modeling: The incorporation of dual-porosity reservoir models, calibrated with crestal well PTA results, is critical for building a robust static and dynamic reservoir characterization. Dual-porosity models can more accurately simulate gas flow by accounting for the fracture network's contribution and the delayed matrix response. History matching should leverage the parameters obtained from the extended well test analysis, including fracture permeability, storativity ratio, and inter-porosity flow coefficient. This modeling approach will enhance production forecasts, optimize field development plans, and support decisions on well spacing, completion strategies, and production optimization.

Conclusion

Extended build-up durations are critical to identifying dual-porosity behavior. The crestal well confirms a naturally fractured system with strong matrix support and the flank wells require longer duration tests to fully characterize fracture influence.

The V-shaped feature on a derivative plot, often referred to as a "V-shaped signature" or "hump" in well test analysis, typically indicates the presence of a fracture or a high-permeability streak in the reservoir. This feature is more pronounced in the derivative curve because it highlights changes in the slope of the pressure transient response, which can be indicative of different flow regimes.

The presence of the V-shaped feature in the derivative plot of the crestal "well A" but its absence in the flank wells "B and C", despite all wells being located in the same dry gas fractured carbonate reservoir, can be attributed to several factors. Variation in fracture density and orientation across the reservoir can lead to differences in permeability, with the crestal area potentially having a higher fracture density or more favorably oriented fractures. Reservoir heterogeneity also plays a role, as fractured carbonate reservoirs are inherently heterogeneous, and the crestal well might be located in a more heterogeneous part of the reservoir.

Well placement and completion can significantly affect the observed well test response, with the crestal well potentially being completed in a way that preferentially accesses more fractured or higher permeability zones. Stress field variations across the reservoir can also impact fracture aperture and permeability, with the crestal area potentially being under a stress regime that enhances permeability. Additionally, variations in fluid saturation across the field, even in a dry gas reservoir, can alter relative permeability and mask the V-shaped feature.

The quality of the well test data and interpretation methodology can also influence the presence or absence of the V-shaped feature. A comprehensive review of the geological setting, well completion details, and specific well test data and analysis methods is necessary to understand the observed differences. Integrating data from other disciplines such as geology, geophysics, and petrophysics can provide a more holistic understanding of the reservoir and variations across the field.

The integration of PTA with complementary methods such as rate transient analysis (RTA), material balance, and full-field reservoir simulation is emphasized as a best practice. This multidisciplinary approach offers a holistic understanding of NCFR behavior, improving decision-making for well placement, stimulation design, and production optimization. This paper underscores the transformative potential of

PTA in naturally fractured carbonate reservoirs, advocating for its inclusion as a core element of reservoir evaluation workflows.

Reference

1. Tiab, D., and Donaldson, E. C. "Petrophysics: Theory and Practice of Measuring Reservoir Rock and Fluid Transport Properties," 3rd ed.
2. Gringarten, A. C. "From Straight Lines to Deconvolution: The Evolution of the State-of-the-Art in Well Test Analysis," SPE- 102079-MS.
3. Kazemi, H. "Pressure Transient Analysis of Naturally Fractured Reservoirs with Uniform Fracture Distribution," SPE-4104-MS.